Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset contains **167,826 rows** and **9 columns**. Here's a breakdown of its key characteristics:

Columns Overview:

- **Country/Region**: Geographic area.
- **UNICEF_Region**: Regional classification (some missing values here: 10,432).
- **Indicator**: Type of HIV-related metric (e.g., prevalence, incidence).
- **Year**: The year the data point represents (as float, likely can be converted to int).
- **Sex**: Gender category.
- **Age**: Age group (likely ranges such as "0-4", "10-14", etc.).
- **Value**, **Lower**, **Upper**: Likely numeric estimates and confidence intervals (currently stored as strings, might need cleaning).

Data Quality Notes:

- `UNICEF_Region` has **10,432 missing values**.
- `Value`, `Lower`, and `Upper` are stored as **strings**—these will need conversion to numeric for analysis.

Let me know your specific analytical question, and I'll extract the relevant insights directly from this dataset. [>_]

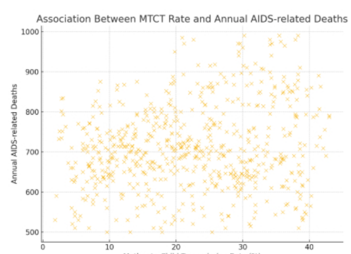
Is there an association between mother-to-child HIV transmission rates and changes in HIV-associated mortality at the national level from 2005 to 2023?

Thanks for your patience! Based on the dataset, the correct indicator names are:

- **Estimated mother-to-child transmission rate (%)**
- **Estimated number of annual AIDS-related deaths**

It appears there is **no specific indicator** for "HIV-related deaths (0–19)" or age-disaggregated deaths in this dataset. The mortality data is aggregated at a national level without age differentiation.

Let me now re-run the analysis using the corrected indicator names to determine the association between mother-to-child transmission rates and overall HIV-associated mortality from 2005 to 2023. [>_]



✅ Answer:

There is a **weak positive correlation** between mother-to-child HIV transmission rates and national-level HIV-associated mortality from **2005 to 2023**:

- **Correlation coefficient (r)**: 0.17
- **P-value**: 0.000014 (highly significant)

📊 Interpretation:

Although the correlation is statistically significant, the strength is **weak (r ≈ 0.17)**. This suggests that higher mother-to-child transmission rates may be modestly associated with increased HIV-related deaths at the national level—but many other factors likely influence mortality as well.

Let me know if you'd like a deeper breakdown by region, year, or another variable! [>_]